CRS-Det: Consistent Region Semantic Modeling for Remote Sensing Open-Vocabulary Object Detection

Ming Liu, Jing Li, Dong Ren, and Hang Sun

***Abstract*—Remote sensing open-vocabulary object detection (RSOVD) aims to detect objects beyond the categories annotated during training. Although vision–language models (VLMs) have significantly advanced this task, existing approaches often suffer from region-level semantic instability, where region representations are highly sensitive to contextual variation and background interference, limiting generalization to novel categories. To address this issue, we propose CRS-Det, which introduces feature-level region semantic modeling to stabilize region representations rather than relying on image-level region discrimination. In particular, we propose multiview region semantic consistency (MRSC) to learn stable region representations by enforcing semantic consistency across multiple views. To reduce the influence of irrelevant background regions, we design region-selective aggregation (RSA), which learns activation-guided spatial weights to emphasize object-relevant visual responses and suppress irrelevant background features. In addition, we introduce prototype-guided semantic alignment (PGSA) to refine textual embeddings using learnable semantic prototypes, improving region–text alignment without introducing image-conditioned prompts. Experiments on four remote sensing benchmarks demonstrate consistent improvements over the controlled baseline and competitive performance against representative open-vocabulary detectors, with particularly notable gains on novel categories.**

***Index Terms*—Open-vocabulary object detection (OVD), region-level semantic, remote sensing image, vision–language models (VLMs).**

I. Introduction

OBJECT detection in remote sensing imagery plays a critical role in earth observation, supporting applications such as environmental monitoring [1], urban planning [2], and disaster response [3]. Over the past decade, detection frameworks have evolved from two-stage designs, such as Faster R-CNN [4], to transformer-based paradigms exemplified by DETR [5], achieving strong performance on large-scale benchmarks. However, most existing detectors [6], [7], [8] operate under a closed-set assumption, where the label space is predefined and fixed. In practical remote sensing scenarios, objects of interest are diverse and continuously emerging, making exhaustive annotation infeasible. This limitation has motivated increasing interest in open-vocabulary object detection (OVD) [9], [10], which aims to recognize both seen and unseen categories using natural language descriptions.

Vision–language models (VLMs) [11], [12], such as CLIP [13], provide a natural foundation for OVD by aligning visual and textual representations in a shared embedding space through large-scale contrastive pretraining. Their ability to define categories via language enables detectors to generalize beyond the training category space. Building on this capability, prior OVD works have explored different strategies to integrate VLMs into detection frameworks. ViLD [14] distills knowledge from a pretrained CLIP [13] into a two-stage detector, transferring open-vocabulary recognition to region-level detection. On the contrary, GLIP [15] unifies detection and grounding by jointly pretraining on large-scale image–text pairs, enabling detectors to exploit textual supervision for both seen and novel categories. These approaches demonstrate the potential of VLMs [16] to bridge conventional closed-set detection and open-vocabulary recognition.

In the remote sensing domain, open-vocabulary detection has recently gained attention due to the complexity of aerial imagery. Small object scales, cluttered backgrounds, and large viewpoint variations make region-level recognition especially challenging. To address these issues, CastDet [17] introduces a teacher–student framework that leverages RemoteCLIP [18] to generate pseudolabels for novel categories. Other representative works, including LLaMA-Unidetector [19], OV-DETR [20], and LAE-DINO [21], further integrate vision–language pretraining or large language models (LLMs) to enhance transferability across base and novel categories. Despite these advances, robust region-level semantic modeling remains an open challenge in remote sensing OVD (RSOVD).

As illustrated in Fig. 1, many VLM-based approaches [17], [19] rely on independently cropped proposals for region-level semantic classification. While cropping isolates the target region, it also removes contextual cues and introduces sensitivity to crop boundaries, leading to unstable region representations. For instance, a groundtrackfield (GTF) may be misclassified as an airport (AP) when surrounding context is partially removed. Conversely, directly using global image representations for classification may introduce excessive scene-level information and weaken region specificity.

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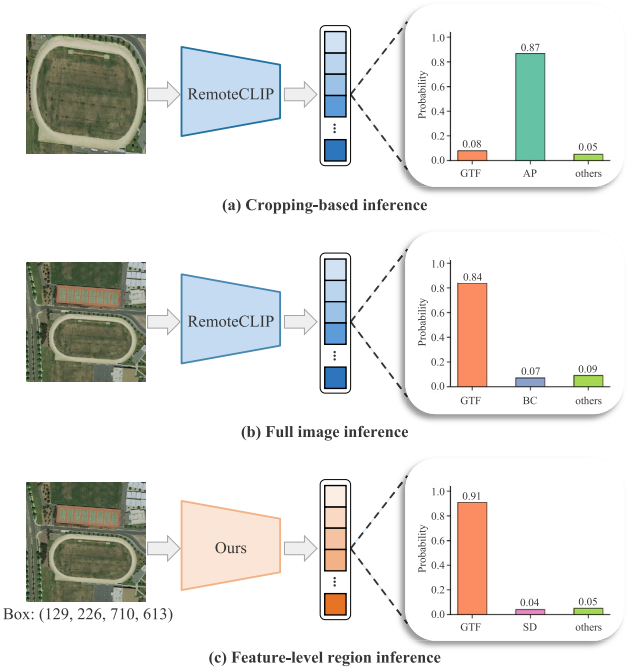


Fig. 1. Comparison of region-level recognition strategies in aerial images. (a) Each proposal is independently cropped from the image and encoded by RemoteCLIP, causing region semantics to be highly sensitive to context removal and crop boundaries. (b) Entire image is encoded to preserve contextual information, but predictions rely on image-level semantics and lack precise region localization. (c) Full image is encoded once, and region representations are derived through consistent region semantic modeling, producing context-aware yet region-focused embeddings for reliable classification. The short names for categories are defined as: GTF, AP, BC, stadium (SD).

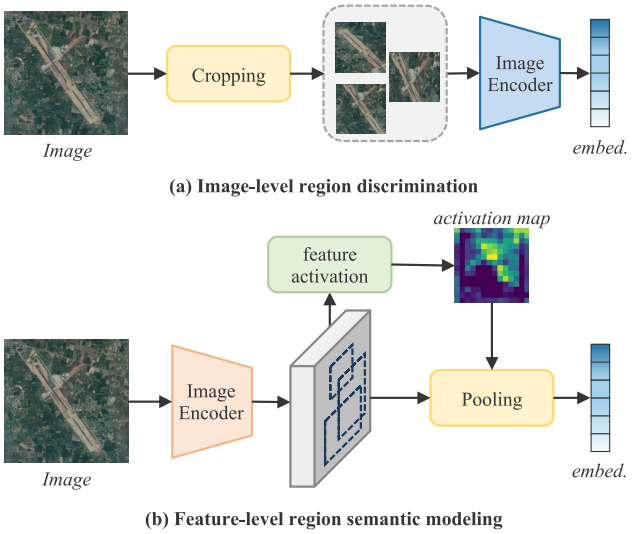


Fig. 2. Comparison of VLM-based region semantic extraction paradigms for open-vocabulary detection. (a) Each proposal is independently cropped and encoded as a separate image for semantic classification. (b) Full image is encoded while spatially corresponding region representations are learned through cross-view consistency and adaptive spatial aggregation (ours).

These observations indicate that region-level semantic instability under contextual variation and background interference is closely related to the performance of RSOVD.

To address this limitation, we propose CRS-Det, which performs feature-level region semantic modeling rather than treating each proposal as an independent image for semantic classification. CRS-Det retains the full-image context during visual encoding and constructs each region embedding from its corresponding spatial visual features, thereby preserving both local object cues and relevant contextual information. In particular, we develop a region-aware vision–language encoder, termed multiview region CLIP (MRCLIP), which incorporates multiview region semantic consistency (MRSC) and region selective aggregation (RSA) to improve region representation stability. MRSC aligns representations of corresponding regions extracted under multiple complementary views, improving semantic robustness under contextual variation. RSA learns activation-guided spatial weights that emphasize discriminative visual responses during region feature aggregation. In addition, we propose prototype-guided semantic alignment (PGSA), which refines textual embeddings through learnable semantic prototypes to improve region–text alignment. As illustrated in Fig. 2, existing approaches [17] typically rely on cropping-based inference, where each region proposal is independently cropped and encoded, making its semantic representation sensitive to context removal and crop boundaries. On the contrary, our approach extracts region representations from full-image visual features and improves their stability through cross-view consistency and adaptive region aggregation.

The main contributions of this article are listed as follows.

1) We identify region-level semantic instability under contextual variation and background interference as an important limitation of existing RSOVD methods, and propose CRS-Det, a feature-level region semantic modeling framework that stabilizes region representations by exploiting spatially corresponding region features from full-image feature maps.
2) We propose MRSC and RSA to enhance visual region representations by encouraging semantic stability across diverse views and reducing the influence of background regions.
3) We develop PGSA to refine textual embeddings through learnable semantic prototypes, improving region–text alignment.
4) Experiments on four remote sensing benchmarks show consistent improvements over the controlled baseline and competitive performance against representative open-vocabulary detectors, with particularly notable gains on novel categories.

II. Related Work

A. Object Detection

Object detection is a fundamental task in computer vision, aiming to localize objects of interest in an image and assign them corresponding categories. With the rapid progress of deep learning, mainstream detectors can be broadly categorized into two-stage and one-stage frameworks [22], [23]. A representative example of the two-stage detectors is Faster R-CNN [4], in which a region proposal network (RPN) first generates candidate boxes that are subsequently refined by classification and regression heads. This design achieves strong accuracy and has served as the foundation for many subsequent approaches. On the contrary, single-stage detectors, such as

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YOLO [24] and RetinaNet [25], streamline the pipeline by directly predicting dense bounding boxes and class scores, striking a balance between efficiency and accuracy. More recently, transformer-based detectors, such as DETR [5], have reformulated detection as a set prediction problem, enabling end-to-end training and eliminating heuristic components like anchors and nonmaximum suppression. Building on this idea, subsequent variants, including Deformable DETR [26] and DINO [27], further improve convergence speed and detection accuracy.

In the field of remote sensing, object detection [28] is critical for large-scale monitoring and analysis. Compared with natural images, remote sensing data pose unique challenges, such as extreme scale variation [29], a high proportion of small objects [30], and densely distributed instances [31], all of which make detection more difficult. Despite remarkable success, conventional detectors remain constrained by a closed-set assumption: they can only recognize a fixed set of categories defined in the training data. This severely limits their scalability in open-world scenarios, particularly in remote sensing, where new object categories frequently emerge. These challenges motivate the exploration of OVD.

B. Open-Vocabulary Object Detection

To overcome the limitations of closed-set detectors, OVD has been proposed. OVD [14], [32], [33] aims to detect objects beyond the predefined training categories by leveraging external semantic knowledge, such as vision–language representations.

Early efforts focus on knowledge distillation [34], where the knowledge of large VLMs such as CLIP [13] is transferred to conventional detectors. Representative works like ViLD [14] and HierKD [35] demonstrate that distilling aligned visual–textual features significantly improves the recognition of novel categories. Another important strategy is to enhance region–text alignment by mapping region features to semantic embeddings with the help of image–text data. OVR-CNN [36] introduces open-vocabulary detection using image–caption pairs to align region features with textual semantics, replacing the traditional classifier with contextualized embeddings. RegionCLIP [37] extends this idea by constructing pseudo region–caption pairs with CLIP and applying contrastive learning, which strengthens local region–text alignment and improves recognition of unseen categories. These studies suggest that beyond proposal feature extraction, maintaining transferable visual–language semantics is crucial for text-defined region classification. More recently, a growing trend is pretraining, where detection is trained jointly with grounding and captioning tasks on massive image–text corpora. Approaches like GLIP [15] and Grounding DINO [38] exemplify this paradigm, pushing OVD toward foundation model capabilities.

Despite these advances, OVD still faces several challenges: accurately recognizing novel categories remains difficult, as region representations are often sensitive to contextual variation and background interference, which consequently affects region–text alignment. These issues are particularly pronounced in remote sensing imagery, where complex backgrounds and large contextual variations exacerbate the difficulty of generalizing to novel categories. Recent attempts have begun to address this gap. For instance, CastDet [17] introduces RemoteCLIP [18] and a dynamic pseudolabel queue to transfer VLM knowledge into aerial object detection, while LLaMA-Unidetector [19] integrates LLMs to unify detection with visual question answering for remote sensing tasks. LAE-DINO [21] introduces dynamic vocabulary construction and visual-guided prompt learning, while OpenRSD [39] proposes a universal open-prompt framework to improve cross-domain generalization. Beyond object detection, recent studies have extended open-vocabulary perception to remote sensing semantic segmentation and instance segmentation by incorporating knowledge transfer, geometric priors, and semantic alignment [40], [41]. However, existing RSOVD methods mainly focus on transferring vision–language knowledge or improving semantic alignment, while the contextual stability of region-level semantics remains underexplored.

Different from existing RSOVD methods, CRS-Det investigates the contextual stability of region-level semantics. Instead of independently encoding cropped proposals, CRS-Det models spatially corresponding region representations from full-image feature maps and improves their robustness through multiview consistency and adaptive region aggregation. This perspective provides a complementary solution for improving region representation stability and region–text alignment in remote sensing open-vocabulary detection.

C. Vision–Language Models

VLMs have demonstrated strong generalization and zero-shot recognition abilities. Representative models such as CLIP [13] and Florence [42] align images and texts within a shared embedding space through large-scale contrastive pretraining, allowing categories to be defined directly by language. Subsequent works, including CoCa [43] and BLIP-2 [44], further extend this paradigm with generative or fusion-based objectives, enhancing the transferability of VLMs across tasks such as classification, retrieval, and segmentation.

To enhance adaptability, several methods refine CLIP-style models via lightweight tuning mechanisms. CoOp [45] introduces learnable prompts to replace handcrafted text templates, while CLIP-Adapter [46] adopts parameter-efficient adapters for downstream adaptation. To address CLIP's limited sensitivity to local details, CLIPSelf [47] employs self-distillation to improve region-level alignment. KAID [48] further explores knowledge-aware interactive distillation to preserve fine-grained visual and cross-modal knowledge in compact VLMs. Related efforts such as DetPro [32] and DeCLIP [49] further enhance robustness through task-aware prompting or multiview contrastive learning.

In the remote sensing domain, these challenges are further compounded by complex backgrounds, large appearance variations, small objects, and diverse spatial structures [30], [50], [51]. RemoteCLIP [18] adapts CLIP [13] to aerial imagery via domain-specific pretraining and provides stronger visual–language representations for remote sensing applications. However, although existing adaptation, self-distillation, and domain-specific pretraining methods improve the transferability of VLMs, they mainly focus on global vision–language alignment, while the stability of region-level semantic representations under contextual variation and background interference remains insufficiently explored. Our

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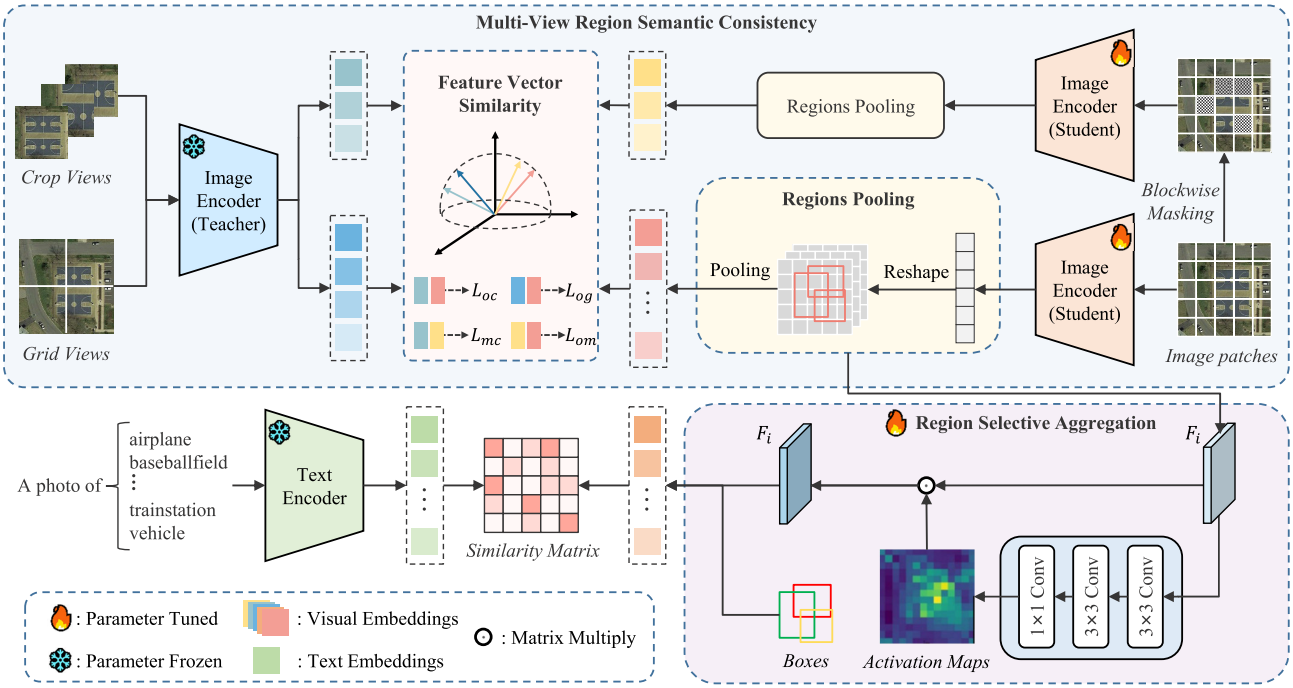


Fig. 3. Overview of the proposed MRSC and RSA.

method addresses this limitation by improving region representation stability and region–text alignment.

III. Methodology

In this section, we formalize the RSOVD problem and provide an overview of the proposed CRS-Det framework. We then introduce the key components for region semantic modeling, including MRSC and RSA, which together enhance region-level visual representations. Building upon these components, we describe a semisupervised detection framework with MRCLIP, followed by PGSA for improving region–text correspondence.

A. Preliminary and Overview

1) Preliminary Definition: RSOVD aims to localize and recognize objects in aerial imagery under an open category set, where the detector must handle categories beyond those annotated during training. Formally, given an image $I \in \mathbb{R}^{H\times W\times 3}$ and a text-specified vocabulary $\mathcal{V}$ provided at inference time (e.g., class names or brief descriptions), the model outputs a set of detections

$$\hat{\mathcal{Y}} = \left\{(b_k, s_k(c)) \;\middle|\; b_k \in \mathbb{R}^4,\ c \in \mathcal{V}\right\} \tag{1}$$

where b_k is the kth bounding box and $s_k(c)$ denotes the alignment score between the region representation and the text representation of class c.

The detector is trained with two sources of data: a labeled dataset whose box annotations cover a base class set $\mathcal{C}_{\text{base}}$, and an unlabeled dataset drawn from a mixture of samples belonging to both base and novel classes $\mathcal{C}_{\text{base}} \cup \mathcal{C}_{\text{novel}}$. No box annotations are available for $\mathcal{C}_{\text{novel}}$. During training, the semantic classifier uses the category names from both $\mathcal{C}_{\text{base}}$ and $\mathcal{C}_{\text{novel}}$, although only the base categories are provided with box-level annotations. At inference, the vocabulary $\mathcal{V}$ may extend beyond $\mathcal{C}_{\text{base}}$, and the model is required to detect both seen (base) and unseen (novel) categories without additional supervision for the novel ones. In addition, category names that are not included in $\mathcal{C}_{\text{base}} \cup \mathcal{C}_{\text{novel}}$ can be introduced after training. These newly introduced categories are directly encoded by the frozen RemoteCLIP text encoder and can be used for region–text similarity computation without retraining the detector.

2) Overview: We propose CRS-Det, a two-phase training framework for RSOVD, which first enhances visual representations and then performs semisupervised detection. In the first stage, we retrain a RemoteCLIP [18] encoder to obtain an improved version termed MRCLIP, which enhances local discriminability while preserving contextual robustness through MRSC and RSA. As illustrated in Fig. 3, MRSC employs two visual encoders, a frozen teacher and a trainable student. The teacher provides stable semantic references by processing region crops and grid-based contextual views, while the student extracts the spatially corresponding region features from the complete image and its block-masked variant. Unlike conventional distillation that primarily transfers logits or features between teacher and student models, MRSC focuses on preserving the semantics of the same spatial region across complementary contextual conditions. Through this multiview consistency learning, MRSC improves region-level robustness to context variation and partial occlusion. RSA further mitigates background interference from global pooling by predicting a semantic activation map within each candidate box and using it to aggregate patch features, leading to region embeddings that focus more precisely on the object itself. The resulting MRCLIP provides context-robust visual features and region-focused representations for subsequent detection.

In the second stage, MRCLIP is integrated into a teacher–student detector to evaluate proposal–text similarity and generate high-confidence pseudolabels for unlabeled data.

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The detector's semantic classifier is further refined through PGSA, which learns semantic prototypes to refine the text embeddings of the base and novel categories included during training, providing stronger semantic anchors for region–text matching.

B. Multiview Region Semantic Consistency

While VLMs such as RemoteCLIP exhibit strong image-level semantic transfer capability, their region-level representations often become unstable once the surrounding context is altered. This problem is particularly pronounced in aerial imagery, where object semantics heavily rely on contextual cues and fine-grained spatial structures. Many remote sensing categories (e.g., airplane, windmill (WM), bridge) require abundant environmental information for correct recognition, while small man-made structures share similar textures. Consequently, removing or perturbing the context often leads to semantic instability, where the same spatial region yields inconsistent embeddings under different contextual conditions, degrading both pseudolabel quality and generalization.

To mitigate this issue, we propose MRSC, a teacher–student consistency learning mechanism designed for region-level semantic stabilization. Different from conventional distillation that mainly transfers knowledge between image-level predictions or features, MRSC performs spatially corresponding cross-view consistency learning, where teacher and student embeddings are extracted from different but aligned visual contexts. The frozen teacher provides reference representations from independently encoded region crops and grid-based contextual views, whereas the trainable student extracts corresponding spatial regions from the feature maps of the full image and its block-masked variant. For each sampled region, the student representation is aligned with the teacher representations derived from the corresponding crop or grid view. This asymmetric multiview design encourages corresponding regions to preserve coherent semantics under context variation, background interference, and partial occlusion.

MRSC employs two parallel encoders that share the same RemoteCLIP backbone architecture but differ in parameter updating strategy. The teacher encoder is frozen at the beginning of training, serving as a stable semantic reference that preserves the pretrained RemoteCLIP embedding space. To construct its inputs, we rely on a pretrained RPN, which is initialized using the pretrained weights provided by CastDet [17], to generate potential object regions from each image. In particular, the generated proposals are first filtered by an objectness score threshold of 0.8, and the top N_p highest-scoring proposals are retained for subsequent feature extraction. The crop set $\mathcal{S}_{\text{crop}} = \{I'_i\}_{i=1}^{N}$, where N corresponds to the number of retained proposals, is obtained by cropping and resizing the original image I according to the generated bounding boxes. In addition, we partition the entire image into an $m \times n$ grid, where both m and n are selected within the range of $[1, 6]$ to capture multiscale contextual variations, and obtain a set of grid-view images by cropping each selected grid region. The grid set is defined as $\mathcal{S}_{\text{grid}} = \{I''_i\}_{i=1}^{M}$, where each I''_i is a grid patch cropped from I. These grid-based patches provide complementary contextual cues, ensuring that both the local object and its surrounding environment contribute to the teacher's semantic representation.

The teacher encoder processes these multiple spatially distinct views and produces a set of region-level embeddings by extracting the [CLS] token from each view

$$\begin{aligned}\mathcal{F}_{\text{crop}} &= \left\{ f_i^{\text{crop}} = \text{CLS}\left(\text{Enc}_{\text{teacher}}\left(I'_i\right)\right) \mid I'_i \in \mathcal{S}_{\text{crop}} \right\} \\ \mathcal{F}_{\text{grid}} &= \left\{ f_i^{\text{grid}} = \text{CLS}\left(\text{Enc}_{\text{teacher}}\left(I''_i\right)\right) \mid I''_i \in \mathcal{S}_{\text{grid}} \right\}. \end{aligned} \tag{2}$$

This produces a semantically stable set of teacher embeddings that serves as supervision for the student. The student encoder is a trainable copy initialized from the same RemoteCLIP weights. It receives two complementary versions of the image: the original image I and a blockwise masked variant I_{mask}, where approximately 30% of the image patches are randomly replaced by mean-value blocks to simulate occlusion and missing context. Both inputs are processed independently, serving as two views that encourage the encoder to learn context-robust and occlusion-invariant representations.

After encoding, the two input images produce sequences of visual tokens T_{ori} and T_{mask}, with $T_{\text{ori}}, T_{\text{mask}} \in \mathbb{R}^{(1+P)\times D}$, where P is the number of patch tokens, D is the embedding dimension, and the first token is the [CLS] representation. We discard the [CLS] token and reshape the remaining P patch tokens back to their spatial grid, forming feature maps $F_{\text{ori}}, F_{\text{mask}} \in \mathbb{R}^{h\times w\times D}$, where $h \times w$ denotes the patch grid resolution of the vision transformer. We define the region set as $\mathcal{R} = \{r_i\}_{i=1}^{K}$, which contains all regions processed by the teacher encoder, including both RPN proposals and selected grid patches. For each region $r_i \in \mathcal{R}$, pool the patch tokens within each region's spatial extent, using the same coordinates as those employed by the teacher, to obtain the student region-level features

$$\begin{aligned}\mathcal{F}_{\text{ori}} &= \left\{ f_i^{\text{ori}} = \text{Pool}\left(\text{Enc}_{\text{student}}\left(I\right), r_i\right) \right\} \\ \mathcal{F}_{\text{mask}} &= \left\{ f_i^{\text{mask}} = \text{Pool}\left(\text{Enc}_{\text{student}}\left(I_{\text{mask}}\right), r_i\right) \right\}. \end{aligned} \tag{3}$$

Each f_i^{ori} and f_i^{mask} is thus aligned to the exact same region r_i used by the teacher, covering both crop and grid views. This ensures that the student embeddings preserve spatial correspondence with the teacher features and facilitate consistent cross-view supervision in the subsequent alignment stage.

To enforce semantic coherence across multiple contextual views, MRSC optimizes the student encoder by minimizing cosine-based discrepancies between region embeddings extracted from complementary teacher–student pairs. As an example, the original–crop consistency loss L_{oc} measures the alignment between the teacher's crop-view embedding f_i^{crop} and the student's original image embedding f_i^{ori}. It is computed as the average cosine similarity loss over all sampled regions

$$\mathcal{L}_{oc} = \frac{1}{N}\sum_{i=1}^{N}\left[1 - \cos\left(f_i^{\text{ori}}, f_i^{\text{crop}}\right)\right]. \tag{4}$$

All embeddings are normalized before similarity computation to stabilize optimization. Analogously, the other consistency terms are defined between different view pairs of the same region: L_{og} aligns f_i^{ori} with f_i^{grid}, L_{om} aligns f_i^{ori} with f_i^{mask}, and L_{mc} aligns f_i^{mask} with f_i^{crop}. The total multiview consistency objective is then formulated as

$$\mathcal{L}_{\text{MRSC}} = \mathcal{L}_{oc} + \alpha_{og}\mathcal{L}_{og} + \alpha_{om}\mathcal{L}_{om} + \alpha_{mc}\mathcal{L}_{mc} \tag{5}$$

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where α_{og}, α_{om}, and α_{mc} are weighting coefficients balancing the four consistency losses.

To further enhance robustness, MRSC applies box-scale augmentation to high-quality regions. Proposals with high scores are randomly enlarged by a factor within the range [1.0, 2.0] before cropping. This strategy encourages the encoder to learn scale-invariant region semantics while remaining sensitive to fine object boundaries. The trained student model, referred to as MRCLIP, preserves the global semantic space of RemoteCLIP while acquiring more stable region representations under varying contextual and spatial conditions.

C. Region Selective Aggregation

MRSC encourages agreement across views but does not specify how region features are pooled from patch tokens. In RemoteCLIP, region embeddings are typically obtained by global (e.g., average) pooling within the bounding box. Such uniform pooling assigns equal weight to all spatial locations, so background and weakly related context are mixed into the representation. As a result, the pooled vector can drift from the object's intrinsic semantics, lowering discriminability and destabilizing alignment with text.

To resolve this, we introduce RSA, a lightweight attention-based module that predicts a semantic activation map for each region and uses it to selectively aggregate the most informative patch features. Rather than fixed, uniform pooling, RSA estimates spatial importance conditioned on the object's semantic response, enabling the region representation to emphasize foreground cues while suppressing irrelevant background. Given the student encoder's feature map from the original image, and a set of ground truth bounding boxes $\mathcal{B} = \{b_i\}_{i=1}^{K}$ provided by the labeled training data, RSA predicts an activation map $A_i \in [0,1]^{h \times w}$ for each region b_i to represent the spatial importance of each patch inside the region. To obtain these activation maps, we first crop the feature map within the box coordinates of b_i and process it through a small convolutional predictor

$$A_i = \text{Softmax}\left(\text{Conv}_{1\times1}\left(\text{Conv}_{3\times3}\left(\text{Conv}_{3\times3}\left(F_i\right)\right)\right)\right) \tag{6}$$

where F_i denotes the cropped feature patch corresponding to region b_i, and the Softmax operation is applied spatially so that all activation values within the region sum to one. This normalization converts the output into a probabilistic attention distribution, allowing RSA to emphasize highly informative foreground patches while suppressing less relevant background areas.

Let $F_i \in \mathbb{R}^{h_i \times w_i \times D}$ be the cropped feature tensor for region b_i, and let $A_i \in [0,1]^{h_i \times w_i}$ be the spatial attention produced by the predictor. The activation-weighted feature map is then denoted as $F_i^{\text{RSA}} = A_i \odot F_i$, where $\odot$ represents elementwise multiplication between the attention map and the feature tensor. The region embedding is then computed by an attention-weighted sum over spatial positions

$$f_i^{\text{RSA}} = \sum_{x=1}^{h_i} \sum_{y=1}^{w_i} A_i\left(x,y\right)\, F_i\left(x,y\right). \tag{7}$$

To train RSA to attend to the correct object regions, we exploit the labeled base data whose categories $\mathcal{C}_{\text{base}}$ are seen during training. Each region embedding f_i^{RSA} is aligned with its corresponding text embedding obtained from the frozen RemoteCLIP text encoder. We normalize all text embeddings and compute the cosine similarity loss, specifically as follows:

$$\mathcal{L}_{\text{RSA}} = \frac{1}{M} \sum_{i=1}^{M} \left[\, 1 - \cos\left(f_i^{\text{RSA}}, \text{Enc}_{\text{text}}\left(\text{prompt}\left(c_i\right)\right)\right)\right] \tag{8}$$

where c_i denotes the ground truth class of the ith region and prompt is the text template "a photograph of a [class]." As training proceeds, RSA gradually learns an internal notion of semantic saliency, which enables the model to emphasize object-related regions while reducing the influence of less relevant background features.

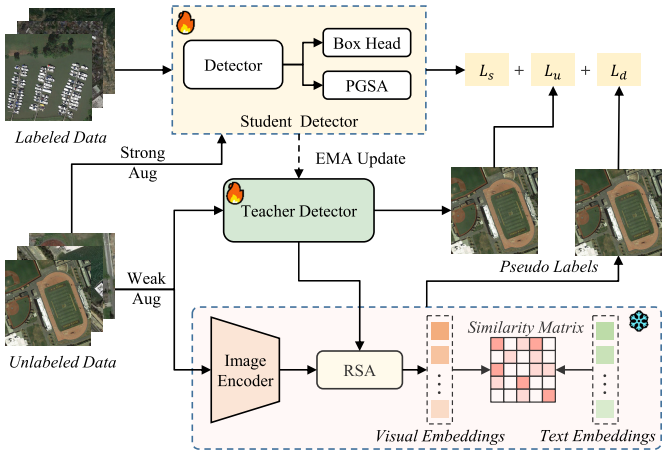


Fig. 4. Integration of the proposed MRCLIP into a semisupervised open-vocabulary detection framework.

D. Semisupervised Detection With MRCLIP

After the MRCLIP encoder is trained in the first stage, we integrate it into a semisupervised detection framework to exploit its enhanced region-level semantics for pseudolabel generation and teacher–student learning. This stage follows the overall architecture of CastDet but employs MRCLIP to generate higher quality pseudolabels for unlabeled data and to promote open-vocabulary generalization. As illustrated in Fig. 4, the framework comprises three key components: a teacher detector, a student detector, and the MRCLIP. The teacher and student share the same detection architecture but maintain separate parameters, while MRCLIP serves as a shared vision–language module that provides semantic guidance for region classification. The teacher's parameters are updated by the exponential moving average (EMA) of the student's weights, serving as a stable target for generating pseudolabels. MRCLIP remains frozen during this stage, serving as a unified vision–language interface that transforms region features into semantic scores.

In each minibatch, B denotes the number of sampled proposals, B^{fg} and B^{bg} the numbers of foreground and background samples, respectively. We denote classification logits by $\mathbf{p}$, class labels by y, predicted boxes by $\tilde{\mathbf{b}}$, and targets by $\mathbf{b}$. We use $\ell_{\text{cls}}(\cdot,\cdot)$ for the classification loss and $\ell_{\text{reg}}(\cdot,\cdot)$ for the regression loss. For labeled samples, the student detector

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is trained using the standard detection loss (classification + regression)

$$\mathcal{L}_{\text{sup}} = \frac{1}{B}\sum_{i=1}^{B}\ell_{\text{cls}}\left(\mathbf{p}_i,\, y_i\right) + \frac{1}{B^{\text{fg}}}\sum_{i=1}^{B^{\text{fg}}}\ell_{\text{reg}}\left(\tilde{\mathbf{b}}_i,\, \mathbf{b}_i\right). \tag{9}$$

For unlabeled images, the teacher detector generates pseudolabels by selecting proposals whose objectness and confidence exceed a threshold, producing pseudoboxes $\mathbf{b}_i^T$ and corresponding class labels $\hat{y}_i^T$. The unsupervised loss consists of classification and regression parts

$$\mathcal{L}_{\text{tea}}^{\text{cls}} = \frac{1}{B^{\text{fg}}}\sum_{i=1}^{B^{\text{fg}}}\ell_{\text{cls}}\left(\mathbf{p}_i,\, \hat{y}_i^T\right) + \frac{1}{B^{\text{bg}}}\sum_{j=1}^{B^{\text{bg}}}\omega_j\,\ell_{\text{cls}}\left(\mathbf{p}_j,\, \hat{y}_j^T\right) \tag{10}$$

$$\mathcal{L}_{\text{tea}}^{\text{reg}} = \frac{1}{B^{\text{fg}}}\sum_{i=1}^{B^{\text{fg}}}\ell_{\text{reg}}\left(\tilde{\mathbf{b}}_i^{\text{fg}},\, \mathbf{b}_i^T\right) \tag{11}$$

where ω_j balances the background term and $\mathbf{b}_i^T$ is the teacher's pseudobox used as the regression target.

To introduce open-vocabulary supervision, each candidate proposal $\mathbf{b}_i$ is evaluated by MRCLIP, which computes the region–text similarity between each region representation extracted from feature maps and all text embeddings. The class with the highest similarity is chosen as the semantic label, and its similarity score serves as the pseudoconfidence. Since these pseudolabels primarily provide semantic supervision, we compute only the classification loss

$$\mathcal{L}_{\text{mr}} = \frac{1}{B}\sum_{i=1}^{B}\ell_{\text{cls}}\left(\mathbf{p}_i,\, \hat{q}_i\right) \tag{12}$$

where $\hat{q}_i$ denotes the class predicted by MRCLIP. Here, $\hat{q}_i$ is selected from the text vocabulary $\mathcal{V}$ provided at the current training stage, allowing MRCLIP to inject semantic supervision for both base and novel categories without introducing additional box annotations. The overall objective is the sum of three parts

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{sup}} + \mu\left(\mathcal{L}_{\text{tea}}^{\text{cls}} + \mathcal{L}_{\text{tea}}^{\text{reg}}\right) + \nu\,\mathcal{L}_{\text{mr}} \tag{13}$$

where μ and ν balance the two unsupervised terms. Through this stage, the enhanced MRCLIP serves as a semantic bridge that not only improves pseudolabel reliability but also enables the detector to generalize to novel categories beyond the supervised base set.

E. Prototype-Guided Semantic Alignment

The semantic classifier projects proposal features into a shared vision–language embedding space, enabling open-vocabulary recognition through text-defined categories. However, the classifier in CastDet relies on fixed text embeddings derived from static prompts (e.g., "a photograph of a [class]"), which remain constant across all images and cannot adapt to variations in visual appearance or context. This causes a *semantic gap* between static textual vectors and dynamically distributed region features, weakening supervision and degrading generalization to unseen categories.

To alleviate this issue, we propose PGSA, which refines text embeddings via cross-attention with a compact set of learnable prototype embeddings, serving as latent semantic bases. Unlike prompt tuning or image-conditioned methods, PGSA performs feature-level adaptation without relying on any image input during inference, acting as a lightweight, learnable prompt layer.

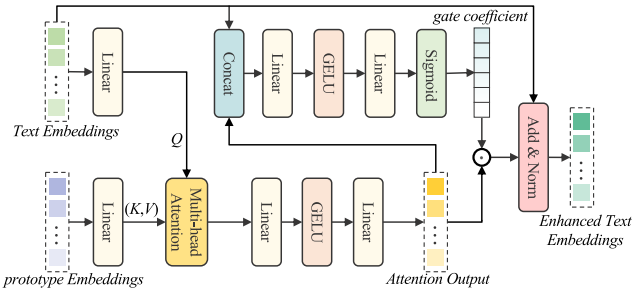


Fig. 5. Architecture of the proposed PGSA.

Formally, let $\mathcal{C}_{\text{train}} = \mathcal{C}_{\text{base}} \cup \mathcal{C}_{\text{novel}}$ denote the semantic category set used during training, and let $C = |\mathcal{C}_{\text{train}}|$. The frozen RemoteCLIP text embeddings of these categories are denoted as $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_C]^\top \in \mathbb{R}^{C\times D}$, where D is the embedding dimension. For each category, the corresponding text prompt is constructed using the standardized template "a photograph of a [class]" and is then encoded by the frozen RemoteCLIP text encoder to obtain $\mathbf{T}$. PGSA introduces learnable prototypes $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_C]^\top \in \mathbb{R}^{C\times D}$, where the number of prototypes is determined by the number of base and novel categories used during training. The prototype bank is optimized during training and remains fixed after training. The prototype embeddings are L_2-normalized at each forward pass to maintain stable attention.

All text embeddings are jointly processed through a multi-head attention block with eight attention heads, where $\mathbf{T}$ serves as the query and $\mathbf{S}$ provides both keys and values

$$\mathbf{Z} = \text{MHA}\left(\mathbf{Q} = \mathbf{T}, \mathbf{K} = \mathbf{S}, \mathbf{V} = \mathbf{S}\right) \tag{14}$$

where $\mathbf{Z} \in \mathbb{R}^{C\times D}$ denotes the attention output having the same dimensionality as $\mathbf{T}$. As illustrated in Fig. 5, PGSA refines the frozen text embeddings by attending to a compact set of learnable semantic prototypes, followed by a feed-forward transformation and gated residual fusion.

The attention output is further refined through a feed-forward transformation and a gating mechanism that adaptively controls the strength of the residual connection. Finally, the enhanced text embeddings are obtained through gated residual fusion followed by L_2 normalization

$$\mathbf{T}' = \text{Norm}\left(\mathbf{T} + \mathbf{G} \odot \mathbf{\Delta}\right) \tag{15}$$

where $\mathbf{T} \in \mathbb{R}^{C\times D}$ is the original text embedding matrix, $\mathbf{\Delta} \in \mathbb{R}^{C\times D}$ denotes the residual features generated by the feed-forward network, $\mathbf{G} \in (0,1)^{C\times D}$ is the gating coefficient controlling the contribution of each dimension, and $\odot$ represents elementwise multiplication. The operation Norm$(\cdot)$ performs L_2 normalization along the feature dimension to ensure that each enhanced embedding has unit length.

During training, the prototype bank $\mathbf{S}$ is optimized jointly with the detection network through the detection training objective, while the RemoteCLIP encoder remains frozen. Through this process, PGSA learns semantic prototypes that refine the textual representations of the base and novel categories included in the semantic vocabulary during training,

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producing textual anchors that align more effectively with proposal features. During inference, PGSA-enhanced text embeddings are used for these base and novel categories. If an additional category $c_{\text{new}} \notin \mathcal{C}_{\text{base}} \cup \mathcal{C}_{\text{novel}}$ is introduced after training, no additional prototype is created. Instead, its text embedding is directly obtained from the frozen RemoteCLIP text encoder and used for region–text similarity computation.

IV. Experiments

In this section, we first describe the datasets and present the implementation details of our framework. We then report comprehensive comparisons with baseline models and representative open-vocabulary detection methods under the same evaluation protocol to assess overall performance. Finally, we conduct ablation studies to verify the effectiveness of each proposed component and analyze their respective contributions.

A. Datasets

1) DIOR [52]: This dataset is a large-scale benchmark for object detection in remote sensing imagery. It contains 23 463 optical images with a spatial resolution ranging from 0.5 to 30 m, covering diverse geographical areas and imaging conditions. The dataset provides over 190 000 annotated object instances belonging to 20 categories. All images are collected from multiple satellite and aerial sensors, exhibiting large variations in scale, illumination, background complexity, and object orientation.

2) VisDroneZSD [53]: This dataset is released for the ICCV Challenge on object detection in drone-captured imagery. It contains 13 067 RGB images with a uniform spatial size of 800×800 pixels. The VisDroneZSD defines 20 object categories, including 16 base categories and four novel categories (AP, basketball court, ground track field, and WM).

3) DOTA [54]: DOTA is a large-scale benchmark for object detection in aerial images and supports both oriented bounding-box and horizontal bounding-box (HBB) evaluation settings. In this work, we use the HBB setting to remain consistent with the standard Faster R-CNN detector adopted in our framework. It contains 2806 high-resolution aerial images with sizes ranging from 800×800 to 4000×4000 pixels and approximately 188 000 object instances across 15 categories.

4) NWPU VHR-10 [55]: This dataset contains 800 very-high-resolution remote sensing images with pixel resolutions between 0.08 and 2 m. It includes ten object categories, with a total of 3651 annotated instances. The dataset was collected from Google Earth and satellite sensors, featuring significant intraclass variation, dense object distribution, and complex backgrounds.

B. Implementation Details

All experiments are conducted on a workstation equipped with an NVIDIA RTX 3090 GPU with 24 GB of memory. Our framework is implemented based on the MMDetection toolbox [57]. We adopt Faster R-CNN [4] as the detection network and use RemoteCLIP [18] as the vision–language backbone. The model is trained using stochastic gradient descent (SGD) with an initial learning rate of 0.01, a momentum of 0.9, and a weight decay of 0.0001. The weighting coefficients α_{og}, α_{om}, and α_{mc} are all set to 1 in our experiments. In addition, the loss weighting factors in (13) are set to $\mu = 2.0$ and $\nu = 1.0$. For the semisupervised learning stage, the confidence threshold for generating high-quality pseudobounding boxes is set to 0.7, and the EMA momentum coefficient for updating the teacher detector is set to 0.999. Unless otherwise specified, the batch size and image preprocessing settings follow those of CastDet [17]. All configurations in the controlled ablation experiments use the same data, detector architecture, vision–language backbone, training settings, and training budget, with the configuration excluding MRSC, RSA, and PGSA corresponding to the CastDet baseline. HM denotes the harmonic mean of mAP_{base} and $\text{mAP}_{\text{novel}}$.

For open-vocabulary evaluation, we follow the class split protocol used in CastDet [17]. In both the DIOR and VisDroneZSD datasets, four categories (AP, basketballcourt (BC), GTF, and WM) are selected as the novel categories, while the remaining 16 are treated as seen classes. For DOTA and NWPU VHR-10 datasets, GTF and BC are designated as novel categories, as summarized in Table I. Note that these datasets may use different naming conventions for the same category (e.g., "GTF" and "ground track field"), but they refer to semantically equivalent classes. Following the semisupervised training protocol of CastDet [17], we additionally incorporate 8725 unlabeled images from the DIOR dataset during training, and adopt the same strategy for the other datasets when unlabeled data are available.

C. Comparison Experiments

To assess the effectiveness of the proposed CRS-Det framework, we compare it with representative open-vocabulary detectors, including ViLD [14], GroundingDINO [38], GLIP [15], OV-DETR [20], BARON [56], LLaMA-Unidetector [19], and CastDet [17], across four remote sensing benchmarks. Tables II–IV summarize the results. Since these methods employ different detector architectures and training paradigms, these comparisons are mainly used to evaluate the overall competitiveness of CRS-Det under the corresponding RSOVD evaluation protocols. The contributions of the proposed components are further isolated through the controlled comparison in Table V, where all configurations use the same detector architecture, data, and training settings. Across all datasets, CRS-Det achieves competitive or improved performance compared with existing methods, with particularly notable gains on novel categories. These results suggest that enhancing region-level representations can contribute to improved open-vocabulary generalization in remote sensing imagery.

1) Comparison on DIOR and VisDroneZSD: Table II presents the comparison on DIOR and VisDroneZSD. On the DIOR dataset, our CRS-Det achieves 63.5 mAP, outperforming the previous best CastDet (61.5 mAP) by a clear margin. More notably, CRS-Det achieves 63.9 $\text{mAP}_{\text{novel}}$, exceeding CastDet and outperforming earlier open-vocabulary detectors such as ViLD, GLIP, and BARON, whose novel-class performance remains substantially lower. These results indicate that incorporating MRSC and RSA is associated with more robust region representations, which benefits similarity-based matching under unseen categories.

A detailed look at categorywise performance further illustrates this advantage: CRS-Det improves high-level semantic classes such as AP and WM, where existing models struggle

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TABLE I

STATISTICS AND CLASS SPLITS OF THE DATASETS USED IN OUR EXPERIMENTS

Dataset	Image size	Train set	Test set	Total categories	Base/Novel Split	Novel categories
DIOR [53]	800	3000	11738	20	16/4	airport, basketballcourt, groundtrackfield, windmill
VisDroneZSD [54]	800	8730	3337	20	16/4	airport, basketballcourt, groundtrackfield, windmill
DOTA [55]	800∼4000	10045	5149	15	13/2	basketballcourt, groundtrackfield
NWPU VHR-10 [56]	∼1000	205	445	10	8/2	basketball court, ground track field

TABLE II

PERFORMANCE COMPARISON ON DIOR AND VISDRONEZSD DATASETS. THE SHORT NAMES FOR CATEGORIES ARE DEFINED AS: AP, BC, GTF, AND WM

Dataset	Method	Detector	AP	BC	GTF	WM	mAP	mAP_{base}	mAP_{novel}	HM
DIOR	ViLD [14]	Faster-RCNN	9.1	11.8	26.7	9.1	45.7	53.5	14.2	22.4
	GroundingDINO [38]	GroundingDINO	22.6	35.6	16.2	20.7	47.2	53.0	23.8	32.8
	GLIP [15]	GLIP	23.7	35.4	15.1	19.1	43.5	48.5	23.3	31.5
	OV-DETR [20]	Deformable DETR	29.1	19.7	25.6	5.2	54.0	62.6	19.9	30.2
	BARON [57]	Faster-RCNN	14.1	16.6	30.4	0.1	50.9	59.8	15.3	24.4
	LLaMA-Unidetector [19]	DINO	18.4	6.8	2.1	4.7	34.6	41.3	8.0	13.4
	CastDet [17]	Faster-RCNN	54.3	78.4	26.3	66.1	61.5	62.8	56.3	59.4
	CRS-Det(Ours)	Faster-RCNN	**59.4**	**82.9**	**45.8**	**67.6**	**63.5**	**63.4**	**63.9**	**63.6**
VisDroneZSD	ViLD [14]	Faster-RCNN	9.1	11.8	26.9	9.1	25.6	28.5	14.2	19.0
	GroundingDINO [38]	GroundingDINO	22.0	36.3	20.4	18.2	28.6	29.7	24.2	26.7
	GLIP [15]	GLIP	26.2	37.3	16.2	20.0	29.6	30.7	24.9	27.5
	OV-DETR [20]	Deformable DETR	29.9	20.5	26.8	5.2	28.7	30.8	20.6	24.7
	BARON [57]	Faster-RCNN	22.8	18.3	34.0	3.0	27.4	29.4	19.5	23.5
	LLaMA-Unidetector [19]	DINO	15.6	6.5	2.5	4.9	24.1	28.3	7.4	11.7
	CastDet [17]	Faster-RCNN	59.9	80.9	31.0	53.7	42.0	**38.4**	56.4	45.7
	CRS-Det(Ours)	Faster-RCNN	**60.8**	**85.0**	**48.4**	**66.7**	**43.3**	37.8	**65.2**	**47.9**

TABLE III

PERFORMANCE COMPARISON ON DOTA DATASETS. THE SHORT NAMES FOR CATEGORIES ARE DEFINED AS: BC AND GTF

Method	Detector	BBD	BD	...	BC	GTF	mAP	mAP_{base}	mAP_{novel}	HM
GroundingDINO [38]	GroundingDINO	38.2	35.4	...	6.2	4.4	48.1	54.7	5.3	9.7
GLIP [15]	GLIP	40.5	27.2	...	12.3	6.3	43.9	49.1	9.3	15.6
LLaMA-Unidetector [19]	DINO	42.4	51.1	...	6.2	2.5	49.6	56.6	4.4	8.2
CastDet [17]	Faster-RCNN	**56.1**	51.3	...	31.6	23.5	57.7	62.3	27.6	38.3
CRS-Det(Ours)	Faster-RCNN	49.9	**51.4**	...	**33.1**	**31.1**	**59.3**	**63.5**	**32.1**	**42.6**

TABLE IV

PERFORMANCE COMPARISON ON NWPU VHR-10 DATASETS. THE SHORT NAMES FOR CATEGORIES ARE DEFINED AS: BC AND GTF

Method	Detector	APL	SH	...	BC	GTF	mAP	mAP_{base}	mAP_{novel}	HM
GroundingDINO [38]	GroundingDINO	68.9	66.2	...	6.9	14.2	47.3	56.4	10.6	17.8
GLIP [15]	GLIP	66.2	63.8	...	6.3	13.7	45.4	54.2	10.0	16.9
LLaMA-Unidetector [19]	DINO	98.8	80.9	...	21.0	5.1	62.3	74.6	13.1	22.3
CastDet [17]	Faster-RCNN	99.4	**89.7**	...	22.4	59.8	76.7	**85.6**	41.1	55.5
CRS-Det(Ours)	Faster-RCNN	**99.6**	89.6	...	**28.4**	**86.2**	**79.7**	85.3	**57.3**	**68.6**

due to background ambiguity in aerial scenes. The observed improvements on both base and novel classes suggest that the proposed representation modeling does not compromise performance on seen categories while improving generalization to unseen ones. Qualitative comparisons with CastDet are presented in Fig. 6. The left and right panels show representative results for base and novel categories, respectively. Compared with CastDet, CRS-Det detects several small or challenging objects that are missed by the baseline results. For some novel-category examples, CRS-Det produces more compact bounding boxes with less background area. These observations visually complement the quantitative improvements reported in Table II.

On VisDroneZSD, CRS-Det achieves the best performance among the compared methods under the same evaluation protocol. It achieves 43.3 mAP and an impressive 65.2 mAP_{novel}, outperforming CastDet and surpassing all open-vocabulary baselines by a significant margin. Drone-captured imagery

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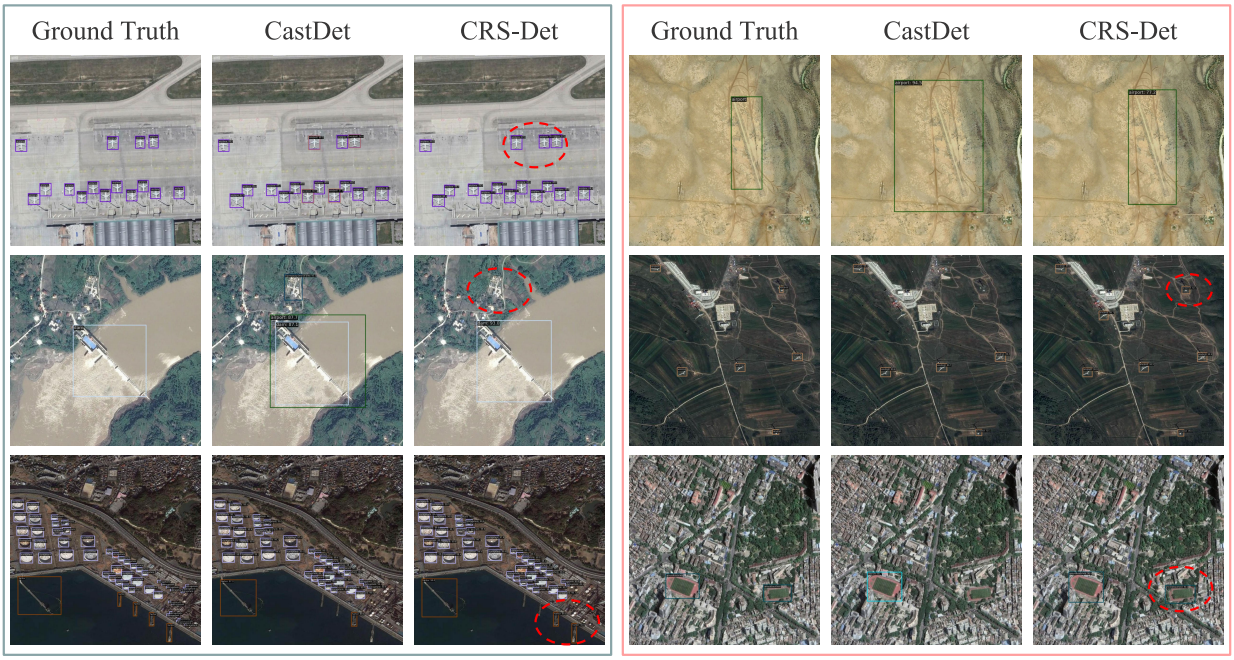


Fig. 6. Qualitative comparison between the ground-truth annotations and the detection results of CastDet and CRS-Det on the DIOR dataset. (Left) representative results for base categories, while (right) results for novel categories. Red dashed circles highlight representative detection differences between CastDet and CRS-Det.

TABLE V
ABLATION EXPERIMENTS OF THE PROPOSED COMPONENTS ON THE DIOR DATASET

MRSC	RSA	PGSA	mAP	mAP_{base}	mAP_{novel}	HM
-	-	-	61.5	62.8	56.3	59.4
✓	-	-	62.7	62.7	62.9	62.8
✓	✓	-	63.1	63.0	63.1	63.0
-	-	✓	62.4	**63.7**	57.0	60.2
✓	✓	✓	**63.5**	63.4	**63.9**	**63.6**

poses challenges such as extreme viewpoints, cluttered backgrounds, and small object sizes. The strong gains delivered by CRS-Det demonstrate that MRSC and PGSA effectively mitigate such difficulties, yielding more stable region representations and improved text–region alignment even under severe perspective distortion. Overall, the results on DIOR and VisDroneZSD clearly show that CRS-Det provides consistent and substantial improvements over existing open-vocabulary approaches, benefiting both base and novel categories and proving particularly effective in complex aerial and drone imagery.

2) Comparison on DOTA and NWPU VHR-10: Table III reports the results on DOTA. CRS-Det achieves the best performance on novel categories, reaching 32.1 mAP, which is a clear improvement over CastDet and all other open-vocabulary baselines. The improvement on novel objects suggests that the learned region representations remain relatively stable under rotations, cluttered backgrounds, and diverse spatial layouts commonly observed in DOTA.

Table IV shows a similar trend on NWPU VHR-10. CRS-Det attains 57.3 mAP on novel classes, substantially outperforming CastDet and indicating improved transferability to unseen categories under the NWPU VHR-10 setting. This improvement highlights the effect of PGSA in refining textual anchors and enhancing region–text alignment. Together, the results on both datasets confirm that CRS-Det consistently offers superior novel-category detection in varied remote sensing scenarios.

D. Ablation Study

To better understand the contribution of each component in our framework, we conduct comprehensive ablation experiments in this part. Our analysis focuses on the overall performance gains brought by the proposed MRSC, RSA, and PGSA modules. In addition, we provide qualitative visualizations to further illustrate how our method improves region representations and enhances semantic alignment in challenging aerial scenes.

1) Overall Effects: Table V presents a controlled comparison based on the same CastDet-style Faster R-CNN framework. The first row, in which MRSC, RSA, and PGSA are all disabled, corresponds to the CastDet baseline reproduced under our experimental setting. All configurations use the same labeled and unlabeled data, detector architecture, RemoteCLIP initialization, training settings, and training budget; therefore, the only variables are the inclusion or exclusion of the proposed components.

Starting from the CastDet baseline, MRSC improves the overall mAP from 61.5% to 62.7% and substantially increases mAP_{novel} from 56.3% to 62.9%. This gain indicates that multiview region consistency is the primary contributor to improved novel-category generalization. Introducing RSA on top of MRSC further improves the overall mAP from 62.7% to 63.1% and mAP_{novel} from 62.9% to 63.1%, demonstrating that activation-guided aggregation provides complementary improvements by reducing background interference during region feature aggregation. When applied independently,

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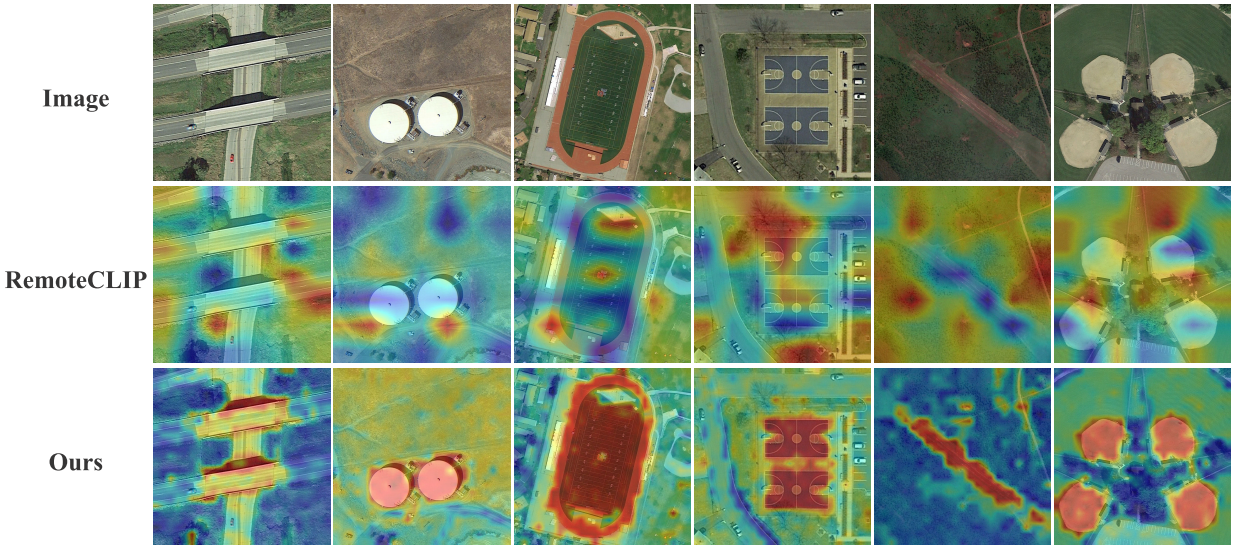


Fig. 7. Visualization of region–text similarity maps. From (top to bottom) input images, similarity maps produced by RemoteCLIP, and similarity maps produced by MRCLIP.

TABLE VI
COMPARISON BETWEEN MP AND THE PROPOSED RSA ON THE DIOR DATASET

Dataset	Strategy	mAP	mAP_{base}	mAP_{novel}	HM
DIOR	MP	62.7	62.7	62.9	62.8
	RSA	**63.1**	**63.0**	**63.1**	**63.0**
DOTA	MP	58.6	63.3	27.4	38.2
	RSA	**58.7**	**63.4**	**28.6**	**39.4**

TABLE VII
ABLATION OF DIFFERENT MRSC CONFIGURATIONS ON THE DIOR DATASET

Settings	mAP	mAP_{base}	mAP_{novel}	HM
baseline	61.5	62.8	56.3	59.4
S1	62.4	**63.1**	59.2	61.1
S2	62.6	62.7	62.1	62.4
S3	62.5	62.6	61.8	62.2
S4	**62.7**	62.7	**62.9**	**62.8**

PGSA improves the overall mAP from 61.5% to 62.4% and the harmonic mean from 59.4% to 60.2%, indicating that refining textual semantic anchors also benefits region–text alignment. When all three components are combined, the model achieves the best results with 63.5% mAP and 63.9% mAP_{novel}. Compared with the CastDet baseline, the complete CRS-Det improves the overall mAP by 2.0 percentage points, mAP_{novel} by 7.6 percentage points, and HM by 4.2 percentage points. These results suggest that the proposed components provide complementary contributions, leading to improved performance when combined.

2) Effects of the RSA: Table VI reports the comparison between standard mean pooling (MP) and the proposed RSA module. On the DIOR dataset, RSA improves the overall mAP from 62.7% to 63.1%, with gains observed on both base and novel categories. This comparison indicates that guiding region aggregation with learned activation maps can yield more discriminative region features than uniform averaging.

3) Analysis of MRSC Design Choices: To understand how different multiview designs contribute to MRSC, we evaluate several configurations in Table VII. First, using only crop and grid views (S1) activates the losses L_{oc} and L_{og}, improving the mAP from 61.5% to 62.4%. This shows that basic cross-view consistency already stabilizes region semantics. Building on S1, introducing multiscale crops in S2 (with the same loss terms) increases the novel-class mAP from 59.2% to 62.1%, indicating improved robustness to geometric variation. Adding random masking in S3 further activates the losses L_{om} and L_{mc} and yields 62.5% mAP, suggesting improved robustness to missing or occluded contextual information.

Finally, S4 integrates all multiview designs and all consistency losses. It achieves the best performance with 62.7% mAP and 62.9% mAP_{novel}, confirming that enforcing consistency across diverse visual conditions is key to learning robust and context-aware region representations.

E. Discussion

1) Visual Analysis: To qualitatively analyze how region semantics are modeled at the feature level, Fig. 7 visualizes the similarity maps between visual features and their corresponding textual embeddings. Compared with RemoteCLIP, which often exhibits scattered and background-sensitive activation patterns, our method produces more compact and object-aligned responses. For challenging aerial scenes containing cluttered backgrounds and elongated structures, RemoteCLIP tends to activate large surrounding regions, indicating strong context dependence. On the contrary, MRCLIP consistently highlights the true object extent with clearer boundaries, suggesting that MRSC contributes to more stable region representations while reducing the influence of irrelevant background regions. These qualitative results provide additional support for the effectiveness of consistent region semantic modeling.

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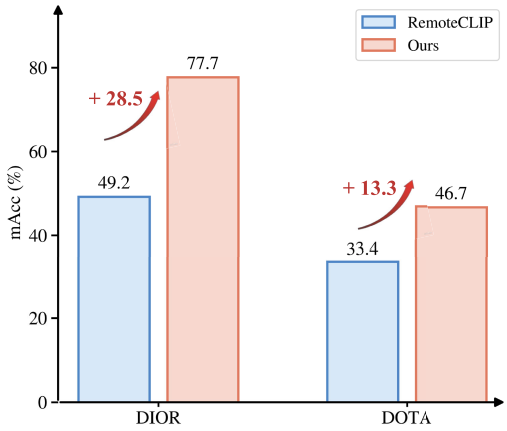


Fig. 8. Region classification accuracy using ground truth bounding boxes. Ground truth regions from different datasets are classified by measuring region–text similarity.

TABLE VIII

REGION CLASSIFICATION ROBUSTNESS UNDER SCALE PERTURBATION AND LOCALIZATION NOISE ON DIOR. THE METRIC IS MACC (%)

Method	Scale Perturbation			Localization Noise		
	0.7	1.0	1.5	0.1	0.2	0.3
RemoteCLIP	30.1	49.2	63.3	48.5	46.9	43.8
MRCLIP	77.4	77.7	77.6	77.6	77.1	76.4

2) Quantitative Analysis of Region Discriminability: To further quantify the stability of region-level semantic representations, we evaluate classification accuracy using ground truth bounding boxes as region inputs. In particular, for each dataset, ground truth regions are extracted and classified by measuring the similarity between region features and text embeddings, without involving any detection heads. As shown in Fig. 8, MRCLIP consistently outperforms RemoteCLIP across different datasets. On DIOR, the classification accuracy improves from 49.2% to 77.7%, while on DOTA, it increases from 33.4% to 46.7%. We further evaluate robustness on DIOR under scale perturbation and localization noise, as reported in Table VIII. RemoteCLIP encodes each perturbed region as an independent crop, whereas MRCLIP extracts the corresponding region representation from the shared feature map. For scale perturbation, the center of each ground truth box is fixed while its width and height are scaled by factors of 0.7, 1.0, and 1.5, thereby changing the proportion of the object, surrounding context, and background within the region. RemoteCLIP varies substantially from 30.1% to 63.3% mAcc, whereas MRCLIP remains stable between 77.4% and 77.7%. For localization noise, the box center, width, and height are jointly perturbed to simulate inaccurate proposals. As the noise level increases from 0.1 to 0.3, RemoteCLIP decreases from 48.5% to 43.8%, while MRCLIP only decreases from 77.6% to 76.4%. These results demonstrate that MRCLIP provides more stable region representations under contextual variation and proposal localization errors.

V. CONCLUSION

This article investigates RSOVD and identifies region-level semantic instability as a key factor limiting generalization to novel categories. We observe that existing approaches predominantly rely on image-level region discrimination, which makes region representations sensitive to contextual variation and background interference in complex aerial scenes. To address this limitation, we propose CRS-Det, which adopts a consistent region semantic modeling perspective to stabilize region representations in the embedding space rather than relying on image-level cropping. Within this perspective, we introduce MRSC to encourage coherent region semantics across diverse views, and RSA to reduce the influence of background regions through activation-guided feature aggregation. In addition, PGSA is employed to refine textual embeddings using learnable semantic prototypes, improving region–text alignment without introducing image-conditioned prompts. Extensive experiments on four remote sensing benchmarks show that CRS-Det achieves consistent improvements over representative baseline methods, with particularly strong gains on novel categories. These results suggest that stabilizing region semantics under contextual variation is particularly important for improving generalization in remote sensing open-vocabulary detection.

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